Contention-Aware Expert Parallel Load Balancing in Mixture-of-Experts Serving in vLLM

Executive Summary

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Abstract

Mixture-of-Experts models distribute sparse expert computation across devices. Expert Parallel Load Balancing periodically changes expert placement to remove runtime hotspots, but applying a placement requires transferring expert weights. In asynchronous mode, these transfers share network resources with live inference. This work formulates the resulting contention problem, develops a lightweight migration-batching algorithm, integrates it into vLLM, and evaluates it on a four-node deployment. In the final A/B experiment, batching improved output throughput by 2.28%, reduced every reported TTFT, TPOT, and end-to-end latency percentile, and reduced head-node NIC P99 by 10.24%. The implementation was reviewed and merged into the upstream vLLM project.

1 Motivation and problem

Sparse MoE layers activate only a subset of their experts for each token, which increases parameter capacity without activating every parameter Shazeer et al. [2017], Fedus et al. [2022]. Under expert parallelism, expert weights are placed on different ranks and tokens are dispatched to the selected experts. The resulting load is input dependent. EPLB observes expert usage and periodically computes a new placement, potentially replicating popular experts.

A synchronous migration pauses inference, so completing all transfers as quickly as possible is the natural objective. Asynchronous EPLB is different: inference continues while weights move in the background. Posting every point-to-point transfer together can create a short NIC burst that interferes with token dispatch, result collection, and other distributed inference communication. The objective is therefore not only to finish migration quickly, but also to protect foreground serving while migration is active.

The offline formulation represents each coalesced source–destination flow p with size w_p. Binary variable $x_{p,b}$ assigns it to batch b, while y_b indicates whether that batch is active. For routed link e, the migration load is

$$L_{e,b} = \sum_p w_p f_{p,e} x_{p,b}. \tag{1}$$

The model first minimizes the worst modeled inference time,

$$T^*_{\text{infer}} = \min T_{\text{infer}}, \qquad T_{\text{infer}} \geq T_0 \left(1 + \frac{L_{e,b}}{C_e(1-\rho)}\right), \tag{2}$$

then minimizes $\sum_b y_b$ while preserving the primary optimum. This strict two-stage order avoids choosing an arbitrary weight between inference protection and batch overhead. In the illustrative eight-rank hotspot, the ILP reduces peak normalized link utilization from 1.50 to 0.50 and modeled inference time from 4.75 to 2.25, using three batches. Figure 1 shows the topology, the unbatched hotspot, and the resulting schedule.

2 Production methodology

The online scheduler avoids a solver and uses deterministic greedy grouping. Remote migrations are first coalesced by ordered rank pair. A flow with endpoints (s, d) has bit mask $(1 \ll s) \vee (1 \ll d)$. It is placed in the first batch whose used-rank mask is disjoint; otherwise a new batch is created. Thus, each batch is a matching in the rank communication graph. Independent rank pairs retain parallelism, while one rank never communicates with several migration peers in the same batch.

The implementation preserves the legacy placement, source assignment, local-copy semantics, and layer activation order. It prepares all layer schedules in one CPU call per rebalance cycle, but expert weights are still transferred

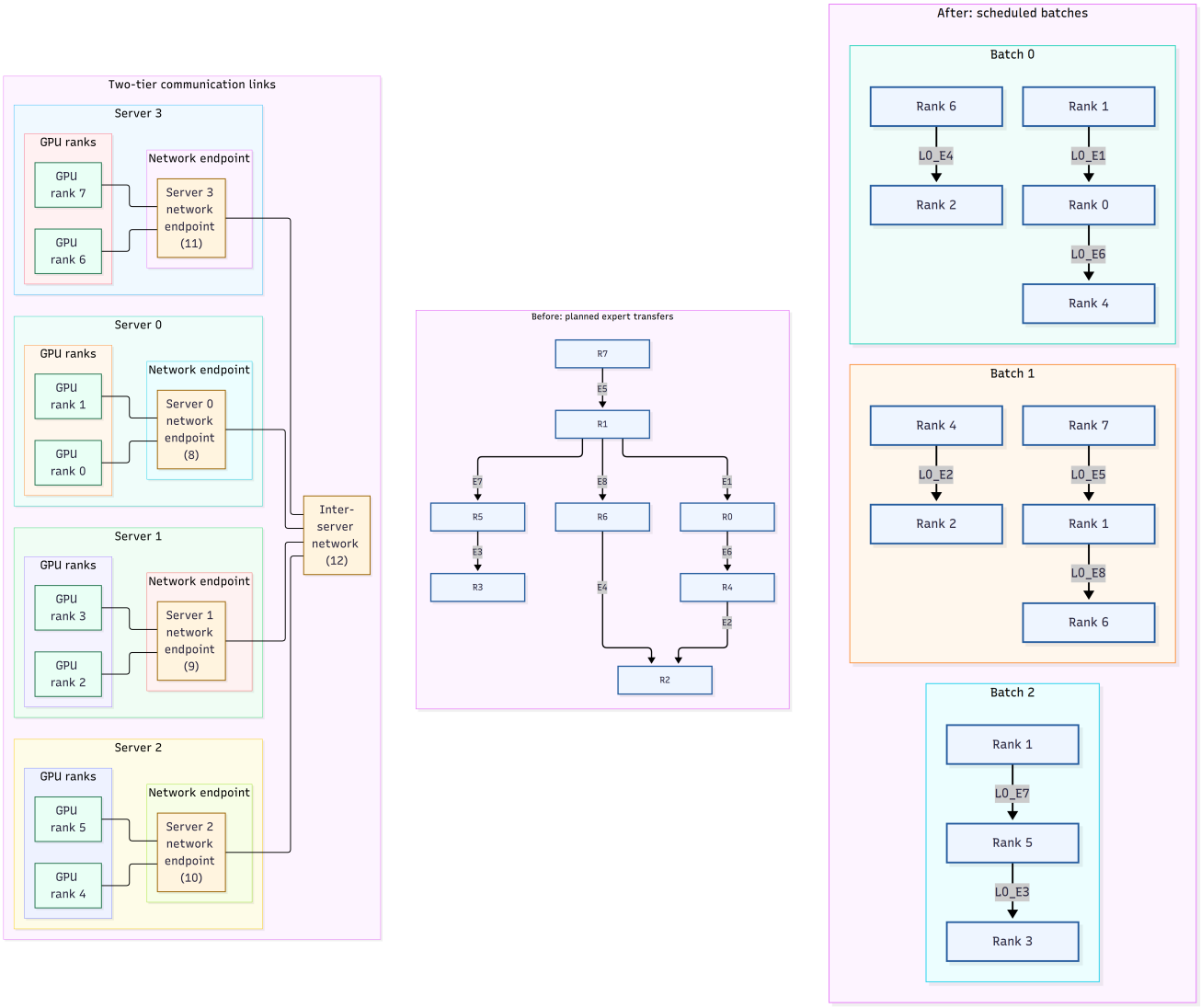


Figure 1: Illustrative topology, unbatched migration hotspot, and the three-batch ILP schedule.

layer by layer. For NIXL, every batch restores the layer transfer context, posts only that batch's sends and receives, and calls the communicator. The feature is opt-in and can operate only with asynchronous EPLB; the synchronous path remains unchanged.

The baseline is the legacy path, which posts all remote transfers for a layer before one communicator execution. The proposed condition changes only the migration-batching configuration flag. This isolates scheduling from placement, model, workload, and backend changes.

3 Evaluation

The final experiment uses four Ubuntu 22.04.5 nodes, one Quadro RTX 6000 GPU per node, and 10 Gb/s Ethernet without RDMA. Qwen3-30B-A3B-Instruct-2507 runs with tensor parallel size four, expert parallelism, Ray, and NIXL over UCX/TCP. The benchmark contains 500 successful requests at concurrency 32, with a 300-token random prefix, 200 additional input tokens, and 300 output tokens. Prefix caching remains enabled. Each condition is one independent run, so the percent changes are descriptive rather than confidence-bound estimates. Table 1 reports the complete comparison, and Figure 2 visualizes the signed improvements.

The NIC metric is obtained independently from one-second RX and TX byte counters on the head node. Its lower P99 supports the intended mechanism, but the study does not attribute individual packets to migration or inference. The workload is synthetic, the off run preceded the on run, and only one cluster and backend are evaluated. These limitations prevent a universal performance claim.

Scheduler overhead is measured separately with NVTX-only Nsight Systems traces. Each range covers schedule construction for all 48 MoE layers. All ranks record 34 cycle-wide calls; the largest cumulative cost is 302.081 ms. The clean A/B run saves 105.932 s, so the conservative cost-to-saved ratio is 0.285%. The profile and serving experiment are separate, so this ratio is a scale comparison rather than a causal decomposition.

Table 1: Final asynchronous A/B result. Latencies are in milliseconds.

Metric	Off	On	Improvement
Output throughput (tok/s)	31.54	32.26	2.28%
TTFT P50 / P99	49180.71 / 75740.25	48599.04 / 74756.17	1.18% / 1.30%
TPOT P50 / P99	828.53 / 1007.82	812.44 / 999.01	1.94% / 0.87%
E2E P50 / P99	303462.69 / 322184.88	293821.76 / 317351.61	3.18% / 1.50%
Head NIC P99 (MB/s)	599.81	538.40	10.24%

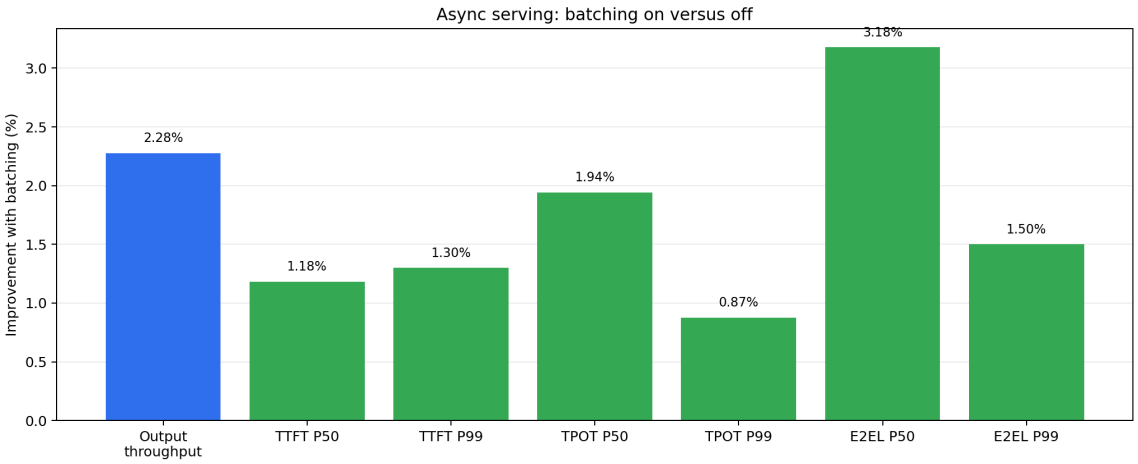


Figure 2: Signed relative improvement with migration batching enabled. Positive values denote higher throughput or lower latency.

4 Conclusions

Asynchronous expert migration should be evaluated as part of serving rather than as an isolated transfer benchmark. The offline model shows why more migration batches can reduce foreground interference even when migration takes slightly longer. The production design turns this observation into a deterministic, dependency-free scheduler with explicit communicator boundaries and a safe opt-in configuration. In the evaluated deployment, throughput, TTFT, TPOT, E2E latency, and NIC tail traffic all improve. Future work should repeat and randomize experiments on server-class GPUs and RDMA fabrics, measure every node, and adapt the per-rank concurrency limit to observed network headroom.

References

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